

WAREHOUSE SLOTTING OPTIMIZER

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

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ABSTRACT

Warehouse efficiency depends on how products are organized within storage locations. Frequently purchased or co-picked items should be placed close together to reduce the travel distance of warehouse workers during order picking. However, determining the optimal placement of products is a complex combinatorial optimization problem, especially in large warehouses with thousands of items. The **Warehouse Slotting Optimizer** is designed to address this challenge by analyzing historical order data to identify product demand and co-pick patterns. Based on this analysis, the system assigns products to appropriate storage slots, minimizing expected travel distance and improving picking efficiency. The project utilizes Data Structures and Algorithms (DSA) such as hash maps for frequency counting, graphs for representing product relationships and warehouse layouts, sorting and priority queues for ranking products, and greedy algorithms for slot allocation. By optimizing product placement, the proposed system enhances warehouse throughput, reduces operational costs, and demonstrates the practical application of DSA concepts in solving real-world logistics and supply chain optimization problems.

Signature of the Guide